

CIVITAS'26

Industry Innovation Challenge Brick PS by Abhishek Bricks

Accelerating Early Strength Development in Fly Ash Brick Manufacturing

Organized by: Concreate Club, IIT Indore

1. Background

Abhishek Bricks is a leading manufacturer of Fly Ash Bricks operating on a semi-automated FAL-G (Fly Ash–Lime–Gypsum) production system at Indore. The plant utilizes a standardized mix design combined with hydraulic pressing, followed by manual handling during stacking and curing operations.

The manufacturing process has been optimized for long-term strength; however, limitations in curing duration, material dependency, and labor intensity have created operational bottlenecks affecting productivity and cost efficiency.

Current Operational Data

A. Primary Mix Composition (Per Batch)

- Stone Dust: 200 kg (~58%)
- Fly Ash: 100 kg (~29%)
- Lime: 30 kg (~9%)
- Gypsum: 11 kg (~3%)

B. Production Cycle

Mixing → Hydraulic Pressing → Manual Stacking → Water Curing (10–15 Days) → Dispatch

C. Key Industrial Challenges

Material Dependency

- Heavy reliance on specific fly ash suppliers, leading to seasonal shortages and supply instability

Extended Curing Duration

- 10-15 day curing cycle results in:
 - High inventory holding
 - Working capital blockage
 - Reduced production turnover

Labor Intensity

- Manual handling required for:
 - Transport of green bricks
 - Multi-level pallet stacking
 - Continuous water curing (Tari)
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2. The Challenge

A multidisciplinary engineering team is tasked with developing a practical, low-cost, and scientifically robust solution to enhance early strength development in fly ash bricks. The proposed solution must be implementable within the existing plant infrastructure, without major capital investment or equipment modification.

Primary Objective: Early Strength Mix Design

Develop an optimized mix design capable of achieving market-saleable compressive strength within 3 days, replacing the conventional 10–15 day curing period.

Scope of Work

The solution must address the following:

- Optimization of proportions of:
 - Fly Ash
 - Stone Dust
 - Lime
 - Gypsum
- Enhancement of:
 - Early hydration kinetics
 - Pozzolanic reaction rate
- Evaluation of chemical accelerators (cost-effective and feasible)
- Exploration of alternative materials, such as:
 - Pond Ash

- Construction & Demolition Waste Powder
- Quarry Dust variants
- Other locally available industrial by-products

Constraints

- Cost per brick must not exceed current production cost
 - No major modification to:
 - Hydraulic press
 - Batching system
 - Solution must be scalable and practical for industrial adoption
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3. Core Technical Domains to Address

Participants are expected to develop an integrated solution covering:

- Early-age strength development mechanisms
 - Reaction kinetics of FAL-G systems
 - Material substitution feasibility and performance
 - Optimization of curing conditions (without infrastructure change)
 - Cost-performance trade-off analysis
 - Durability and long-term performance considerations
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4. Key Questions to Be Explicitly Addressed

The proposal must clearly answer:

- How can early compressive strength be enhanced within 72 hours using the existing mix system?
 - What material modifications or substitutions can reduce dependency on specific fly ash sources?
 - What role can chemical accelerators play, and what are their cost implications?
 - How does the proposed mix impact long-term strength and durability?
 - Can curing efficiency be improved without altering infrastructure?
 - What are the economic benefits in terms of reduced curing time and inventory turnover?
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5. Evaluation Criteria

Criterion	Weightage	Description
Technical Feasibility	40%	Practical applicability and scientific validity
Cost Analysis	30%	Cost per brick and potential savings
Strength & Quality	20%	Compressive strength and durability performance
Innovation	10%	Use of waste materials and creative solutions

6. Deliverables

Mandatory Submission

Each team must submit:

1. A 2-page Technical Proposal (PDF) including:

- Proposed solution
- Technical justification
- Preliminary cost analysis
- Expected strength performance

2. On-Campus Brick Casting Requirement

- Teams will be required to visit the IIT Indore campus for on-site casting of their proposed brick mix design under provided facilities and supervision.
- Teams must arrive 3 days prior to the event to complete the casting process and necessary preparations for testing.

3. Compressive Strength Testing

- Physical samples of the proposed mix design may be required.
 - Compressive strength testing will be conducted on the event day to evaluate early strength performance and validate the proposed solution.
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7. Expected Outcomes

- Reduction in curing duration from 10–15 days to ≤ 3 days
 - Faster inventory turnover and improved cash flow
 - Reduced dependency on limited raw material sources
 - Lower labor intensity and improved operational efficiency
 - Scalable and industry-ready innovation
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8. Code of Conduct

- Maintain discipline and professionalism throughout the event
 - Respect fellow participants and organizers
 - Judges' decisions will be final and binding
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9. Available Materials

- Ordinary Portland Cement (OPC)
- Fly Ash
- Ground Granulated Blast-furnace Slag (GGBS)
- Silica Fume
- Rice Husk Ash (RHA)
- Lime
- Stone Dust
- Quarry Dust
- Polypropylene (PP) Fiber
- Micro Synthetic Fiber
- Gypsum / Phosphogypsum